# Unlock the code: F = ma

13 C

**DIRECTIONS** · Net force first! Copy the letter into the lock.

### STATION 1

A 2 kg cart: 6 N right, 2 N left. Find a.

(V) 2 m/s² right (F) 4 m/s² right

### STATION 2

A 4 kg cart: 12 N right, 4 N left. Find a.

(E) 2 m/s² right (I) 4 m/s² right

### STATION 3

A 3 kg cart: 5 N right, 5 N left. Find a.

(C) 0 m/s² (S) 3.3 m/s²

### STATION 4

A 4 kg box accelerates 2 m/s² right; friction 6 N left. Applied force?

(T) 14 N (A) 8 N (B) 2 N

### STATION 5

18 N right, 6 N left, a = 3 m/s² right. Mass?

(O) 4 kg (U) 6 kg

### STATION 6

2 kg: 2 N right, 10 N left. Find a.

(R) 4 m/s² left (L) 6 m/s² right

**LOCK** [1] [2] [3] [4] [5] [6]

Hint: net force is a ___ sum of all the forces.

R7 (1) (2) (3)

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

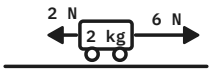
UNIT 1 · DAY 13 · R7 · SEI · PARTNER TALK

# Talk it through: net force first

Language goal: I can **explain** each step of F = ma.

**DIRECTIONS** · Partner A and B read their lines aloud. Listen, then write. Switch each round.

## ROUND 1 · 6 N RIGHT, 2 N LEFT, 2 kg CART

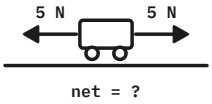



**A ASKS** "What is the net force on the cart?"

**B SAYS** "I subtract: ____ − ____ = ____ N, pointing ____."

**A CHECKS** "So the acceleration is ____ ÷ ____ = ____ m/s². I agree / disagree because ____."

## ROUND 2 · SWITCH · 5 N RIGHT, 5 N LEFT

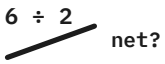



**B ASKS** "Is the cart accelerating?"

**A SAYS** "The net force is ____, so a = ____. The cart keeps a ____ velocity."

**B ADDS** "I want to add that zero net force does not mean ____."

## ROUND 3 · BOTH · A MISTAKE: a = 6 ÷ 2




**A** "The student used only the ____ force."

**B** "The fix is to find the ____ force first, so a = ____."

### LISTENING LOG · MY PARTNER SAID…

### NOW I WRITE IT MYSELF

To use F = ma, I first…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R7 (1) (2) (3)

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# Four steps, every F = ma problem

13 M

**DIRECTIONS** · Draw the forces, find the net force, then divide by mass.

**PROBLEM 1** · A 5 kg box is pulled 20 N right while friction is 5 N left. Find its acceleration.

**1 READ**

m ____ · forces ____ · find ____

**2 DRAW**

free-body sketch

**3 SOLVE**

$\Sigma F_x = F_{right} - F_{left} = ma$

**4 WRITE**

The box accelerates ____ m/s² ____ because the net force is ____ .

**PROBLEM 2** · A 4 kg box accelerates right at 2 m/s² while friction is 6 N left. Find the applied force.

**1 READ**

m ____ · a ____ · friction ____ · find ____

**2 DRAW**

free-body sketch

**3 SOLVE**

$F_{applied} - 6 = (4)(2)$

**4 WRITE**

The applied force is ____ N, more than ma because ____ .

Hardest box: ☐ Read ☐ Draw ☐ Solve ☐ Write

R7 1 2 3

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# Fix the group chat

13 R

**DIRECTIONS** · Reply in the empty bubbles. Show the math on the right.

**Physics study group**

**ELI**

5 kg box, 20 N pull, 5 N friction. a = 20 ÷ 5 = 4 easy

**YOU ①**

**MAZ**

3 kg cart, 5 N right 5 N left. a = 0 so it has to be stopped

**YOU ②**

**NADIA**

2 kg: 2 N right, 10 N left. which way does it accelerate?

**YOU ③**

**ELI**

how do I find mass if I know the forces and a?

**YOU ④**

**SHOW YOUR RECEIPTS**

① net force first

② a = 0 → ?

③ ΣF_x with signs

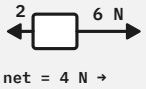
④ m = ΣF ÷ a

R7 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Picture dictionary: F = ma

Language goal: I can **explain** net force.



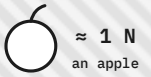

## net force ΣF

fuerza neta · força resultante

all forces combined

my language: _____

use it: _____



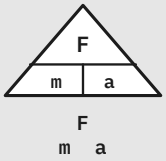

## newton N

newton · newton

the unit of force

my language: _____

use it: _____




## F = ma

F = ma · F = ma

force = mass × accel.

my language: _____

use it: _____



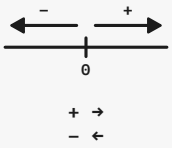

## opposite

opuesto · oposto

pointing the other way

my language: _____

use it: _____



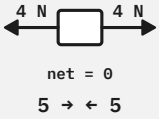

## direction

dirección · direção

which way it points

my language: _____

use it: _____




## balanced

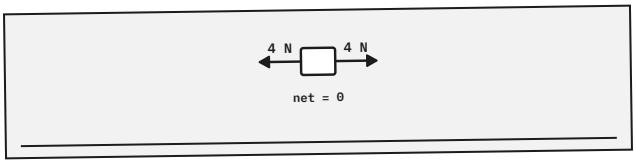
equilibrado · equilibrado

net force = 0

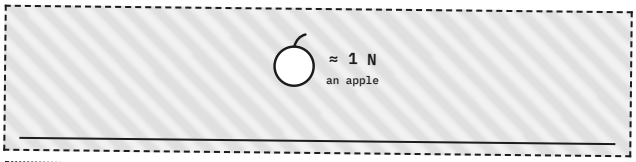
my language: _____

use it: _____

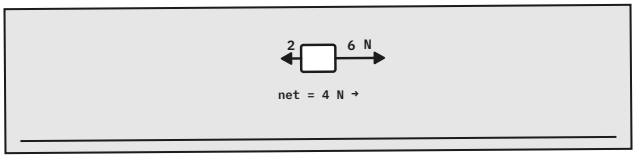
### MATCH · WRITE THE WORD FOR EACH PICTURE



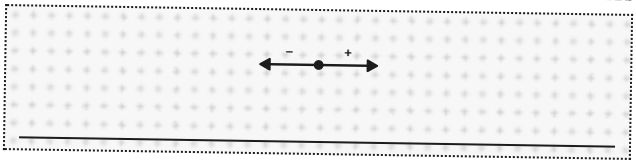

_____




_____




_____




_____

I can point and say all 6 words to a partner ☐

R7 (1) (2) (3)

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# I have... who has? F = ma

13 G

**DIRECTIONS** · Start at ★. Solve, find the tile that "has" your answer, and keep going.

**★ START** — **A**

WHO HAS…

5 kg box: 20 N right, 5 N left. Net force?

I have **0** — **B**

WHO HAS…

A 4 kg box at 2 m/s², friction 6 N. Pull?

I have **3 m/s² right** — **C**

WHO HAS…

2 kg: 8 N right, 2 N left. a?

I have **4 kg** — **D**

WHO HAS…

1 N equals…?

I have **15 N right** — **E**

WHO HAS…

That box's acceleration?

I have **1 kg·m/s²** — **F**

WHO HAS…

Acceleration points the same way as…?

I have **3 m/s²** — **G**

WHO HAS…

3 kg with 5 N each way. a?

I have **14 N** — **H**

WHO HAS…

18 N right, 6 N left, a = 3. Mass?

I have **the net force** — **I**

**END**

**MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END**

R7 1 2 3

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# Comic: the net force decides

13 CS

**YOUR READER**
A 6th grader who thinks the bigger team in tug-of-war "just wins."

Stick figures are fine. The physics gets checked.

**MUST-HAVES · check it off, then color its circle**

- ☐ ORANGE — Force arrows sized to match 60 N and 40 N
- ☐ PENCIL — "net force = ____ N, toward ____" in panel 2
- ☐ GREEN — The motion in panel 3 (speeding up)

**1 · THE PULL** *Team A pulls 60 N left. Team B pulls 40 N right.*

CAPTION _____

**2 · FIND THE NET** *A character figures out the net force.*

CAPTION _____

**3 · SO IT ACCELERATES** *The rope and teams speed up toward one side.*

CAPTION _____

**EXPLAIN IT ·** In 1–2 sentences, what physics does your comic show? Use: **net force · F = ma · acceleration · direction**

R7 1 2 3

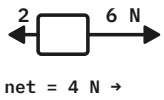
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# Doodle Notes · net force

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

**Net force** · fuerza neta (ES) · força resultante (PT)




**TODAY'S CASE**

A 2 kg cart feels 6 N right and 2 N left.

## 01 · SKETCH AND LABEL

Sketch both orange force arrows to scale on the same cart.

## 02 · CHANGE THE PICTURE

Draw the single equivalent net-force arrow in a separate box. Add the acceleration magnitude and direction.

**Annotate your sketches.** Circle the mark that best shows net force. Add a label in English or your home language.

**SAY IT TO A PARTNER · THEN WRITE**

I subtract these force magnitudes because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
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# Frayer Model · net force

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

**Net force** · fuerza neta (ES) · força resultante (PT)

**A** · A push on the cart and its reaction on a hand cancel on the cart.

**B** · Two opposite pushes on one cart combine to a net force.

### 01 · DEFINITION

What does net force mean in your own words? Use the notes or start in your home language.

### 02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

### 03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

### 04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

**SAY IT TO A PARTNER · THEN WRITE**

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else